UNIVERSITY OF CENTRAL PUNJAB

(Faculty of ITCS)

Operating Systems (OS)

Assignment No 1 **Date: 14 Oct 2025**

Total Marks: 100 **Due Date: 21 Oct 2025**

Assignment Guidelines:

*Assignment should be **handwritten** using **A4 Page**, no torn papers from registers will be accepted.*
*Assignment is to be done **individually** and should be submitted at the **start of class**.*
*Answers should be **to the point** & in your **own words**; **no copying** from any source.*
Late submission will cause you to lose 25% per day; after three days of deadline no submissions will be entertained.

CLO 1 Questions

1. What is an interrupt. Why are interrupts important for CPU efficiency. List and explain different types of interrupts with real-world examples.
2. What benefit is achieved by performing interrupt driven I/O compared to polling based I/O. Under what specific conditions would polling be considered more efficient.
3. Describe the mechanism used for enforcing memory protection in order to prevent a program from modifying the memory associated with another program. Draw relevant diagrams too.
4. Describe the mechanisms used by an OS to enforce memory protection, ensuring that a program cannot modify the memory space of another program. Support your explanation with relevant diagram showing how memory protection is implemented.
5. Consider a scenario where a malicious or errant user-mode program attempts to violate system integrity. Explain how the CPU dual mode of operation and privileged instructions prevent this violation. Also discuss what hardware support is provided by the system to implement this protection.
6. Differentiate between multiprogramming & multitasking. What benefit is achieved in time-sharing system.
7. Differentiate between a System Call and an API call (via a system library function) from both a programmer's and an OS kernel's perspective. Explain how a single standard C library function, like printf(), may require multiple, distinct system calls to accomplish its task.
8. Explain the difference between protection and security in the context of operating systems. List and briefly describe the mechanisms used by the OS to implement protection and security.
9. Evaluate the Microkernel structure's claim to greater extensibility and reliability compared to a Monolithic kernel. What are the key performance overheads inherent in a microkernel design.
10. The strictly layered approach is rarely used in commercial OSs today. Identify the primary performance and design inflexibility issues that led to its obsolescence.
11. What are Loadable Kernel Modules (LKMs) used in modern operating systems like Linux. Discuss how they combine the advantages of monolithic and microkernel designs, with reference to extensibility, performance, reliability, and ease of maintenance.
12. Explain the difference between asymmetric clustering and symmetric clustering in the context of high availability. Briefly describe the problem of maintaining data consistency and coherence in clustered systems. How can data consistency and coherence be achieved in such systems.
13. In what ways can parameters be passed from a user program to the kernel during a system call.

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